

Smart Eye Protection and Alert System

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Abstract— *This paper presents the design and development of a Smart Eye Protection and Alert System aimed at promoting healthy screen usage habits and reducing eye strain, fatigue, and potential health risks associated with prolonged exposure to digital screens. The system is built using an Arduino Uno microcontroller and integrates multiple sensors to monitor user behaviour and environmental conditions in real time. A Passive Infrared (PIR) sensor is used to detect the presence of a user, while an ultrasonic sensor continuously measures the distance between the user and the screen to ensure a safe viewing range. Additionally, dual Light Dependent Resistor (LDR) sensors are employed to compare ambient light with screen brightness, enabling the system to detect improper lighting conditions that may cause discomfort or eye strain.*

The collected sensor data is processed by the microcontroller using intelligent decision-making logic, allowing the system to identify unsafe conditions such as excessive proximity to the screen, imbalance in lighting, or prolonged inactivity. A built-in timer tracks the duration of screen usage, and a sleep detection mechanism is implemented to identify user inactivity over extended periods (e.g., 10 minutes), which may indicate drowsiness or fatigue. Based on these conditions, the system provides real-time feedback through an LCD display and a buzzer with distinct sound patterns for different alerts, such as “Too Close,” “Adjust Light,” and “Wake Up.”

The proposed system operates automatically without requiring manual intervention, making it suitable for students, professionals, and individuals who spend long hours in front of screens. It can also be beneficial in educational environments, workplaces, and home settings to encourage healthier digital habits. The use of low-cost components, efficient sensor integration, and real-time monitoring ensures that the system is both affordable and reliable. Overall, the Smart Eye Protection and Alert System demonstrate an effective and practical solution for enhancing user awareness, improving visual ergonomics, and preventing long-term health issues associated with improper screen usage.

Keywords— *Eye protection, Arduino Uno, ultrasonic sensor, PIR sensor, LDR, distance monitoring, light detection, sleep detection, embedded systems, human-computer interaction, smart health system, real-time monitoring, automation, alert system, ergonomics.*

I. INTRODUCTION

The rapid advancement in technology and the widespread use of digital devices have significantly increased human interaction with screens in daily life. Devices such as computers, laptops, tablets, and smartphones have become essential tools in education, professional work, and communication. However, excessive and improper usage of these devices has led to various health-related issues, particularly eye strain, fatigue, and reduced visual comfort. Continuous exposure to screens without maintaining proper distance, lighting conditions, and rest intervals can negatively affect both physical health and productivity.

The integration of major engineering domains such as electronics, embedded systems, and automation has enabled the development of intelligent systems capable of monitoring human behaviour and environmental conditions in real time. These systems reduce human effort and provide a safer and more efficient working environment. With the help of sensors and microcontrollers, it is now possible to design cost-effective solutions that can automatically detect unsafe conditions and provide immediate feedback to the user.

One of the primary challenges in maintaining healthy screen usage habits is the lack of awareness and real-time monitoring. Users often sit too close to screens, work under improper lighting conditions, or remain inactive for extended periods without taking necessary breaks. Such practices can lead to eye strain, headaches, reduced concentration, and long-term health issues. Therefore, there is a need for a smart system that continuously observes user activity and alerts them when unsafe conditions are detected.

This paper presents the design and implementation of a **Smart Eye Protection and Alert System** using an Arduino Uno microcontroller. The system integrates multiple sensors to monitor user presence, distance, ambient lighting, and inactivity. Based on the collected data, the system processes information in real time and provides alerts through an LCD display and a buzzer, ensuring improved user awareness and healthier screen usage habits.

A. Distance and Presence Monitoring

To achieve effective monitoring of user presence and position, the system utilizes a Passive Infrared (PIR) sensor along with an ultrasonic distance sensor connected to the Arduino Uno. The PIR sensor detects the presence of a user by sensing motion, while the ultrasonic sensor continuously measures the distance between the user and the screen.

The ultrasonic sensor operates using high-frequency sound waves, which are transmitted into the environment and reflected back after hitting an object. The time taken for the echo to return is used to calculate the distance. This non-contact method provides accurate and real-time distance measurements. If the user moves too close to the screen, the system detects the unsafe distance and generates an alert to maintain a safe viewing range.

Additionally, the system tracks variations in distance over time to determine whether the user is still present, even when no motion is detected by the PIR sensor. This combined approach improves reliability and ensures continuous monitoring of user position.

B. Brightness Management System

To achieve proper brightness management in the system, Light Dependent Resistors (LDRs) are used in combination with fixed resistors to form a voltage divider circuit, which is connected to the analog input pins of the Arduino Uno. This configuration enables the system to measure light intensity in an analog form rather than a simple digital signal.

The LDR is a photo-resistive component whose resistance varies inversely with the intensity of incident light. When the surrounding light intensity increases, the resistance of the LDR decreases, and when the light intensity decreases, its resistance increases. By connecting the LDR in series with a fixed resistor, a voltage divider is formed that produces a variable output voltage corresponding to the light level. This analog voltage is then read by the Arduino's built-in Analog-to-Digital Converter (ADC), which converts it into a digital value ranging from 0 to 1023.

In the proposed system, two LDR-based voltage divider circuits are used to monitor ambient lighting and screen brightness conditions. The Arduino continuously reads the analog values from both sensors and compares them to determine the balance between environmental light and screen illumination. If the difference between the two readings exceeds a predefined threshold, the system identifies it as an improper lighting condition.

Based on this analysis, the system generates an alert to prompt the user to adjust the lighting environment or screen brightness. This helps in maintaining optimal visual conditions and reduces the risk of eye strain. The use of an analog sensing approach provides higher accuracy and sensitivity compared to digital LDR modules, allowing the system to detect even small variations in light intensity and respond effectively.

C. Sleep Detection and Inactivity Monitoring

To achieve sleep detection and inactivity monitoring in the system, sensor data from the ultrasonic sensor and motion

detection from the PIR sensor are continuously analyzed over time using the Arduino Uno microcontroller. The system is designed to monitor user activity by observing changes in position and movement patterns.

The ultrasonic sensor is used to measure the distance between the user and the screen at regular intervals. Any significant change in distance indicates user movement, while minimal or no change over a period of time suggests inactivity. The PIR sensor complements this by detecting initial motion, allowing the system to identify when a user is present.

The microcontroller maintains a timer that tracks the duration since the last detected movement. If no significant variation in distance is observed for a predefined time interval, such as 10 minutes, the system assumes that the user is inactive or possibly drowsy. This condition may occur when the user falls asleep or remains idle for an extended duration.

Upon detecting prolonged inactivity, the system generates an alert to notify the user. This feature helps in preventing fatigue, maintaining alertness, and encouraging active engagement during screen usage. The implementation of inactivity monitoring enhances the overall functionality of the system by addressing not only physical posture but also user behaviour over time.

D. Alert and Rest Reminder System

To achieve effective user notification and promote healthy screen usage habits, the system incorporates an alert and rest reminder mechanism using an LCD display and a buzzer. These components provide both visual and auditory feedback based on the conditions detected by the sensors.

The Arduino Uno processes the data obtained from the ultrasonic sensor, PIR sensor, and LDR-based brightness system to determine whether the user is maintaining safe and comfortable conditions. Depending on the detected scenario, the system displays appropriate messages such as "Too Close," "Adjust Light," and "Wake Up" on the LCD screen.

In addition to visual alerts, a buzzer is used to generate different sound patterns corresponding to different conditions. For example, rapid beeping is used for unsafe distance, slower beeping for improper lighting, and a continuous tone for inactivity or sleep detection. This differentiation in sound patterns allows the user to quickly identify the type of alert without needing to constantly observe the display.

Furthermore, the system can be extended to include periodic rest reminders, encouraging users to take short breaks after prolonged screen usage. Continuous exposure to screens without breaks can lead to eye strain and reduced productivity. By providing timely alerts and reminders, the system promotes better work habits and improves overall user well-being.

II. LITERATURE SURVEY

In recent years, significant research has been carried out in the field of smart health monitoring systems and human-computer interaction, particularly focusing on reducing the negative effects of prolonged screen usage. Various approaches have been proposed using embedded systems, sensors, and

automation techniques to improve user safety, comfort, and awareness.

Several studies have explored the use of **ultrasonic sensors** for distance measurement and safety monitoring. Ultrasonic-based systems are widely used due to their non-contact nature, low cost, and reliable performance in real-time applications. Researchers have implemented these sensors in applications such as obstacle detection, assistive devices, and posture monitoring systems. These systems demonstrate the effectiveness of ultrasonic sensing in maintaining safe distance and preventing harmful conditions.

The use of **Passive Infrared (PIR) sensors** has also been extensively studied for motion detection and presence monitoring. PIR sensors are commonly used in security systems, automatic lighting, and occupancy detection due to their simplicity and low power consumption. However, many studies highlight the limitation of PIR sensors in detecting only motion and not static objects. To overcome this limitation, researchers have proposed combining PIR sensors with other sensing techniques, such as ultrasonic or infrared sensors, to improve accuracy and reliability.

In the domain of **light intensity monitoring**, Light Dependent Resistors (LDRs) have been widely used due to their simplicity and cost-effectiveness. Many systems utilize LDRs to control brightness levels automatically in applications such as smart lighting, energy-saving systems, and environmental monitoring. However, most earlier implementations rely on digital threshold-based detection, which provides limited accuracy. Recent approaches emphasize the use of analog LDR configurations with voltage divider circuits to achieve more precise measurement of light intensity, enabling better decision-making in brightness control systems.

Research has also been conducted on **fatigue detection and inactivity monitoring systems**, particularly in areas such as driver safety and workplace monitoring. These systems often use sensors, cameras, or wearable devices to detect signs of drowsiness or prolonged inactivity. While camera-based systems provide high accuracy, they are often expensive and computationally complex. Sensor-based approaches, on the other hand, offer a cost-effective and simpler alternative for detecting inactivity by analyzing motion patterns and behavioural changes over time.

Additionally, several studies have focused on **alert and notification systems** to improve user interaction and response. These systems use visual displays, sound signals, or vibration mechanisms to inform users about unsafe conditions. The use of distinct alert patterns has been shown to improve user response time and understanding, especially in real-time monitoring systems.

Despite these advancements, many existing systems address only a single parameter, such as distance, lighting, or motion, rather than providing a comprehensive solution. There is a lack of integrated systems that simultaneously monitor multiple factors affecting eye health and user well-being. Furthermore, cost and complexity remain significant challenges in the widespread adoption of advanced monitoring systems.

The proposed **Smart Eye Protection and Alert System** aims to address these limitations by integrating multiple sensors, including ultrasonic, PIR, and analog LDR-based brightness detection, into a single embedded platform. By combining distance monitoring, lighting analysis, inactivity detection, and real-time alert mechanisms, the system provides a more comprehensive and efficient solution for promoting healthy screen usage and preventing eye strain.

III. PROBLEM STATEMENT

- (a) Prolonged usage of digital devices without proper monitoring leads to eye strain, fatigue, and reduced visual comfort.
- (b) Users often maintain an unsafe distance from the screen, which increases stress on the eyes and may result in long-term vision problems.
- (c) Improper lighting conditions, such as imbalance between ambient light and screen brightness, cause additional discomfort and reduce viewing efficiency.
- (d) Lack of awareness regarding continuous screen usage without breaks leads to inactivity, drowsiness, and decreased productivity.
- (e) Existing systems generally focus on a single parameter and fail to provide a comprehensive solution for monitoring distance, lighting, and user activity simultaneously.
- (f) Absence of real-time feedback prevents users from correcting unsafe viewing habits effectively.
- (g) There is a need for an automated and cost-effective system that can monitor user behavior and environmental conditions and provide timely alerts to prevent health-related issues.

IV. PROPOSED SYSTEM

The proposed system is a Smart Eye Protection and Alert System designed to monitor user behaviour and environmental conditions during screen usage and provide real-time feedback to promote healthier practices. The system is developed using an Arduino Uno microcontroller as the central processing unit, which integrates multiple sensors including a Passive Infrared (PIR) sensor, an ultrasonic distance sensor, and Light Dependent Resistors (LDRs) configured in a voltage divider circuit for analog light measurement. The PIR sensor is used to detect the initial presence of a user, while the ultrasonic sensor continuously measures the distance between the user and the screen to ensure a safe viewing range. The LDR sensors monitor ambient lighting and screen brightness levels, enabling the system to detect improper lighting conditions that may cause eye strain.

The system also incorporates a timer mechanism to track the duration of continuous screen usage and an inactivity detection feature to identify prolonged periods without movement. Based on the processed sensor data, the system provides feedback

through a Liquid Crystal Display (LCD) and a buzzer. Different alert messages and sound patterns are generated to indicate specific conditions such as unsafe viewing distance, lighting imbalance, or user inactivity. The proposed system is designed to be cost-effective, easy to implement, and suitable for use in educational, professional, and home environments. By integrating multiple sensing techniques and real-time decision-making, the system provides a comprehensive solution for improving visual ergonomics and reducing the risk of eye-related health issues.

V. METHODOLOGY

The working of the proposed system is based on continuous monitoring of user behaviour and environmental conditions using multiple sensors, followed by real-time processing and feedback through output devices. Initially, the Passive Infrared (PIR) sensor detects the presence of a user by sensing motion. Once motion is detected, the system becomes active and begins tracking the user's position using the ultrasonic sensor. The ultrasonic sensor emits high-frequency sound waves, which reflect from the user and return to the receiver. The time taken for the echo to return is used to calculate the distance between the user and the screen. This distance is continuously compared with a predefined safe threshold to determine whether the user is positioned too close.

To ensure accurate presence detection even when the user is stationary, the system stores the last measured distance at the time of motion detection and continuously compares it with the current distance. If the variation in distance exceeds a predefined limit, the system assumes that the user has moved away; otherwise, it considers the user to still be present. This approach overcomes the limitation of the PIR sensor, which can only detect motion and not static objects.

For brightness management, Light Dependent Resistors (LDRs) are used in a voltage divider configuration to provide analog measurements of light intensity. The Arduino reads the analog values corresponding to ambient light and screen brightness through its Analog-to-Digital Converter (ADC). These values are continuously compared to determine the balance between environmental lighting and screen illumination. If the screen brightness is significantly lower than the ambient light level, the system identifies this condition as "Brightness Low" and generates an alert. Similarly, if the screen brightness is excessively higher than the surrounding light, the system identifies it as "Brightness High" and prompts the user to adjust the lighting conditions. This classification provides more precise feedback compared to simple mismatch detection.

The system also incorporates inactivity monitoring to detect possible fatigue or drowsiness. It continuously tracks the time since the last detected movement using a timer. If no significant movement or distance variation is observed for a predefined duration, such as ten minutes, the system interprets this as user inactivity and triggers a sleep alert. This helps in maintaining user alertness and preventing prolonged inactivity.

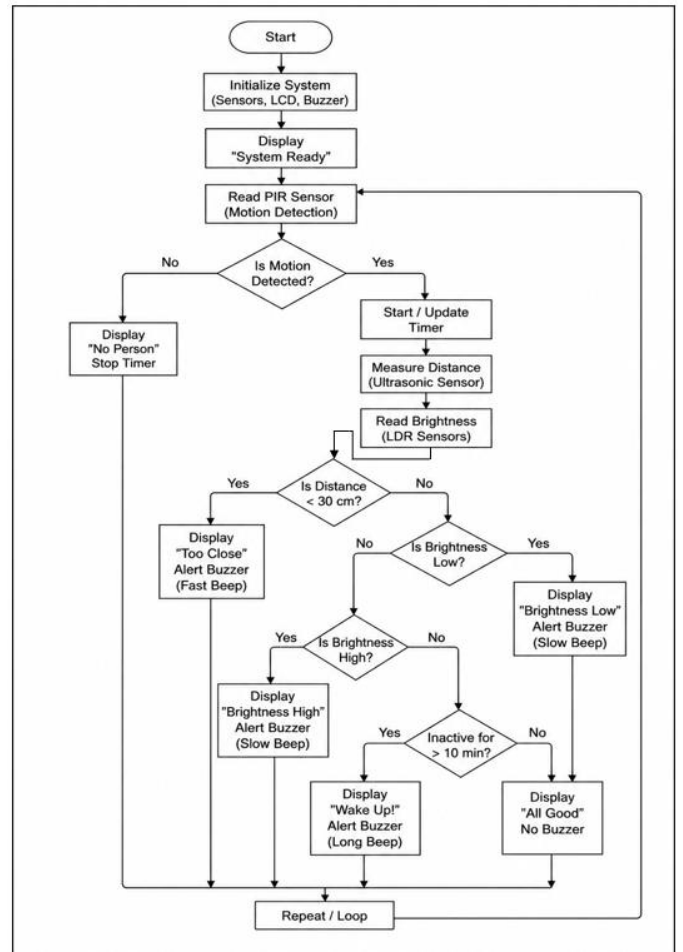


Fig. 1 Flowchart of Smart Eye Protection System

Based on the processed data, the Arduino generates appropriate outputs through an LCD display and a buzzer. The LCD provides real-time messages such as "Too Close," "Brightness Low," "Brightness High," and "Wake Up," while the buzzer produces distinct sound patterns corresponding to different alert conditions. This combination of continuous sensing, intelligent processing, and real-time feedback ensures effective monitoring and promotes safe and healthy screen usage habits.

VI. HARDWARE IMPLEMENTATION

A. Arduino UNO

The Arduino Uno is an open-source microcontroller board developed by Arduino.cc, which serves as the main control unit of the system. It consists of digital and analog input/output pins that can be interfaced with various sensors and external devices. The board has 14 digital input/output pins, out of which 6 can be used for PWM output, and 6 analog input pins for reading analog signals.



Fig. 1 Arduino UNO

The Arduino Uno is programmed using the Arduino Integrated Development Environment (IDE) via a USB cable. It can be powered either through the USB connection or by an external power supply ranging from 7V to 12V. In this system, the Arduino processes the data received from sensors such as PIR, ultrasonic sensor, and LDRs, and controls the output devices like the LCD and buzzer.

B. PIR Sensor

The Passive Infrared (PIR) sensor is used for detecting the presence of a user by sensing motion. It works by measuring changes in infrared radiation emitted by objects in its surroundings. When a human body moves within its detection range, the sensor outputs a HIGH signal, indicating motion detection.



Fig. 2 PIR Sensor

The PIR sensor is connected to a digital input pin of the Arduino. It plays a crucial role in activating the system when a user is detected and helps in initiating further monitoring

processes such as distance measurement and inactivity tracking.

C. PIR Sensor

The ultrasonic sensor is used to measure the distance between the user and the screen. It operates using ultrasonic sound waves with frequencies above 20 kHz, which are beyond human hearing. The sensor consists of a transmitter and a receiver.

The transmitter emits ultrasonic pulses, which travel through the air and reflect back upon hitting an object. The receiver captures the reflected waves, and the time taken for the echo to return is used to calculate the distance. This sensor is connected to the Arduino using trigger and echo pins.

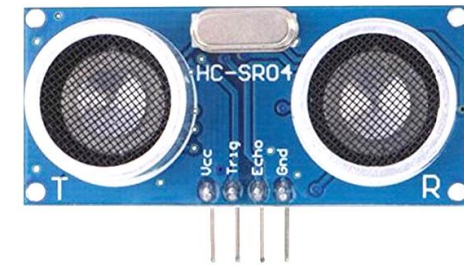


Fig. 3 Ultrasonic Sensor

The ultrasonic sensor helps in ensuring that the user maintains a safe viewing distance from the screen and also assists in detecting presence when the user remains stationary.

D. LDR (Light Dependent Resistor)

A The Light Dependent Resistor (LDR) is used to measure light intensity in the environment. In this system, two LDRs are used along with fixed resistors to form voltage divider circuits, which provide analog output corresponding to light intensity. The resistance of the LDR decreases with an increase in light intensity and increases when the light intensity is low. The voltage output from the divider circuit is given to the analog input pins of the Arduino, where it is converted into digital values using the ADC.



Fig. 4 LDR Sensor

One LDR is used to measure ambient light, while the other is used to measure screen brightness. The comparison of these values helps in determining whether the brightness is too low or too high, and appropriate alerts are generated.

E. LCD Display (16×2 with I2C Module)

A 16×2 Liquid Crystal Display (LCD) is used to display system status and alert messages. The LCD is interfaced with the Arduino using an I2C module, which reduces the number of required connections to just two communication lines: Serial Data (SDA) and Serial Clock (SCL).

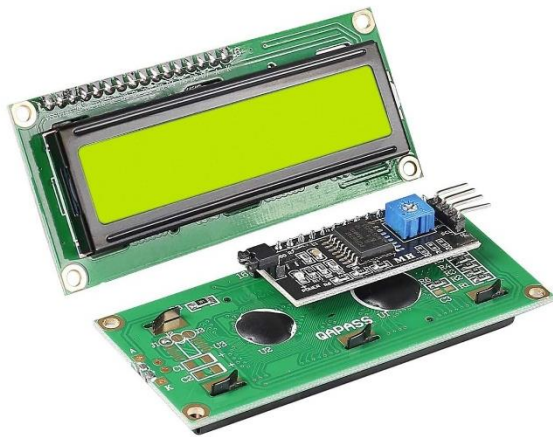


Fig. 5 LCD Display

The LCD displays real-time messages such as “Too Close,” “Brightness Low,” “Brightness High,” “Wake Up,” and system status messages. This provides visual feedback to the user regarding their screen usage conditions.

F. Buzzer

The buzzer is used as an audio output device to generate alerts. It is connected to a digital output pin of the Arduino. The buzzer produces different sound patterns using the `tone()` function, allowing differentiation between various alert conditions.



Fig. 6 Buzzer

For example, fast beeping is used for unsafe distance, slow beeping for brightness issues, and continuous sound for sleep detection. This helps the user identify alerts without constantly observing the display.

G. Hardware Implementation of the model

The complete hardware setup involves connecting all sensors and output devices to the Arduino Uno. The PIR sensor is connected to a digital input pin, while the ultrasonic sensor is connected through trigger and echo pins. The LDRs are connected to analog input pins through voltage divider circuits. The LCD is connected using the I2C interface, and the buzzer is connected to a digital output pin.

All components share a common ground and are powered through the Arduino board. The proper arrangement and connection of components ensure smooth functioning and accurate data acquisition.

VII. VIRTUAL SIMULATION

The virtual simulation of the proposed Smart Eye Protection and Alert System is carried out to verify the functionality and

behaviour of the system before implementing it on hardware. Simulation tools such as Proteus or similar platforms are used to design and test the circuit by integrating the Arduino Uno microcontroller with virtual components including sensors, LCD display, and buzzer.

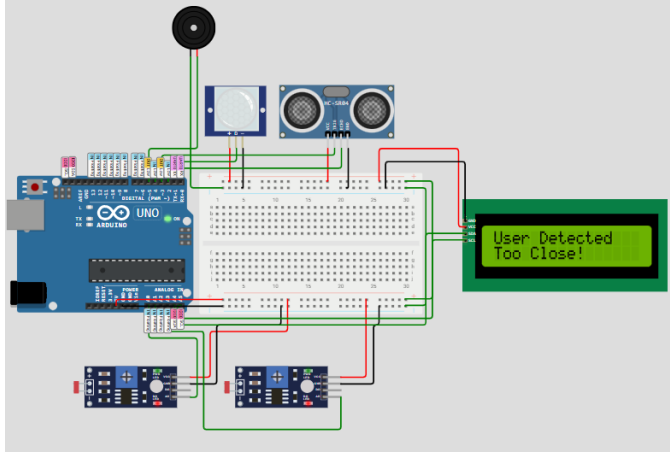


Fig. 7 Virtual Implementation

In the simulation environment, the connections between the Arduino and the components are established according to the designed circuit diagram. The program developed in the Arduino IDE is compiled and uploaded into the simulated Arduino board. The behaviour of sensors such as the ultrasonic sensor and LDRs is tested by varying input parameters, which represent changes in distance and light intensity.

The simulation allows observation of system responses under different conditions. For example, when the simulated distance value falls below the threshold, the system displays a “Too Close” message on the LCD and activates the buzzer. Similarly, when there is a mismatch in brightness levels, the system displays “Brightness Low” or “Brightness High.” Inactivity conditions are also simulated to verify the sleep detection mechanism, where the system generates a “Wake Up” alert after a predefined duration.

The virtual simulation helps in identifying design errors, verifying logic implementation, and ensuring proper working of the system before physical deployment. It reduces the risk of hardware damage and improves the overall reliability of the system.

VIII. WORKING PROTOTYPE

The working prototype of the Smart Eye Protection and Alert System is developed by integrating all hardware components on a physical setup. The Arduino Uno is used as the central controller, connected to the PIR sensor, ultrasonic sensor, LDR-based brightness detection circuit, LCD display, and buzzer.

When the system is powered on, it initializes and displays a ready message on the LCD. The PIR sensor detects the presence of a user and activates the system. Once activated, the ultrasonic sensor continuously measures the distance between the user and the screen. If the user moves too close to the screen, the system generates an alert through both the LCD display and the buzzer. Simultaneously, the LDR sensors measure ambient light and screen brightness. Based on the comparison of these values, the system identifies whether the brightness is too low or too high and displays appropriate messages such as “Brightness Low” or “Brightness High,” along with an alert sound.

The system also tracks user inactivity by monitoring changes in distance over time. If no significant movement is detected for a predefined duration, such as ten minutes, the system assumes inactivity or drowsiness and triggers a “Wake Up” alert. Different buzzer patterns are used to indicate different conditions, allowing easy identification of alerts.

The prototype demonstrates the real-time performance of the system, showing its ability to monitor multiple parameters simultaneously and provide immediate feedback. The successful implementation of the prototype validates the effectiveness of the proposed system in promoting safe and healthy screen usage.

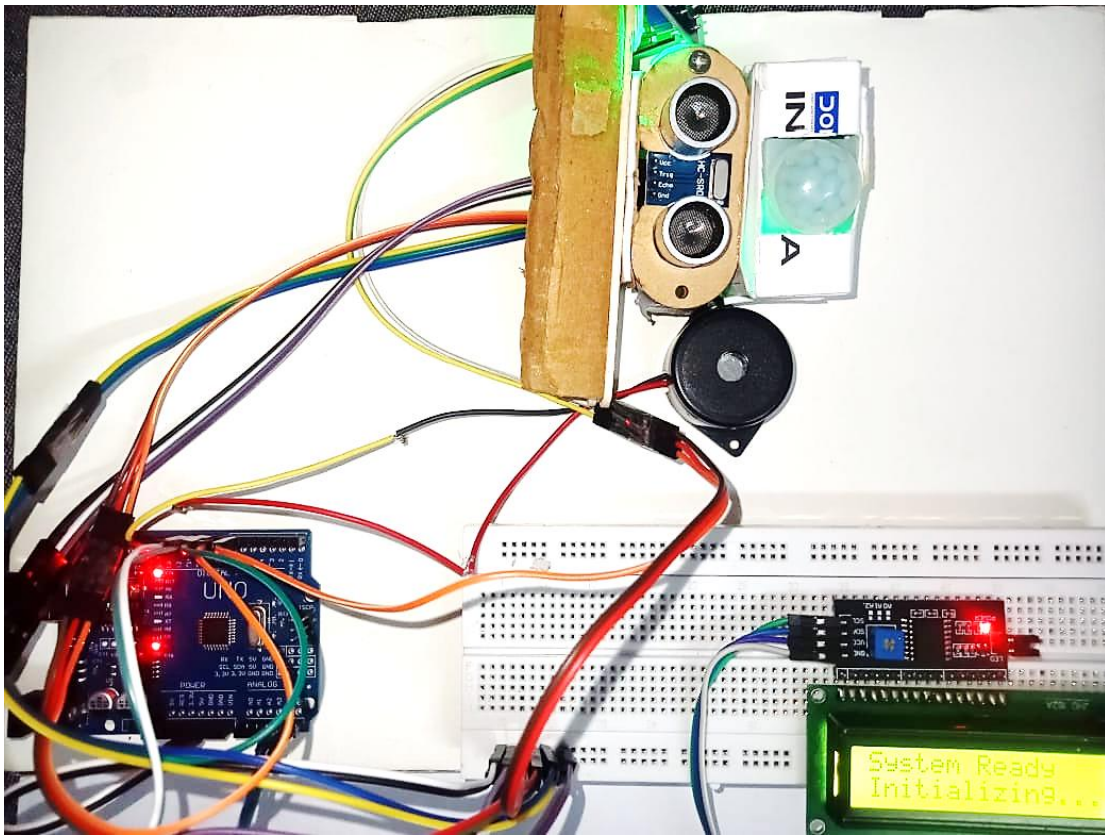


Fig. 8 Working Prototype(Full System)

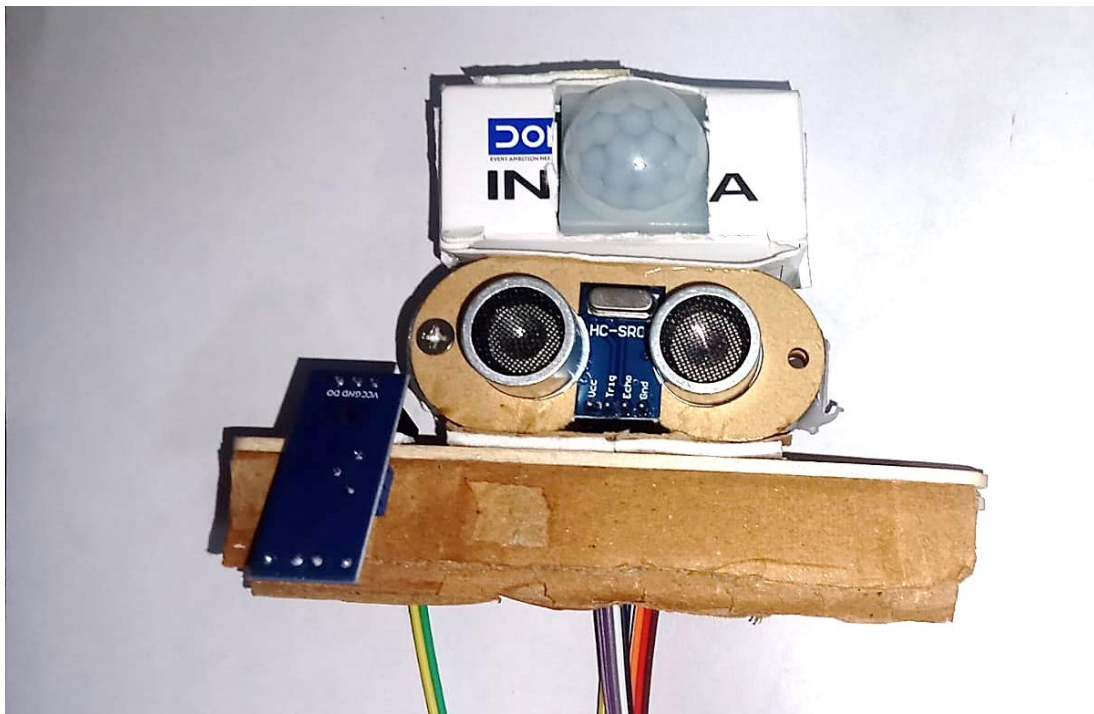


Fig. 9 Working Prototype(Head Only)

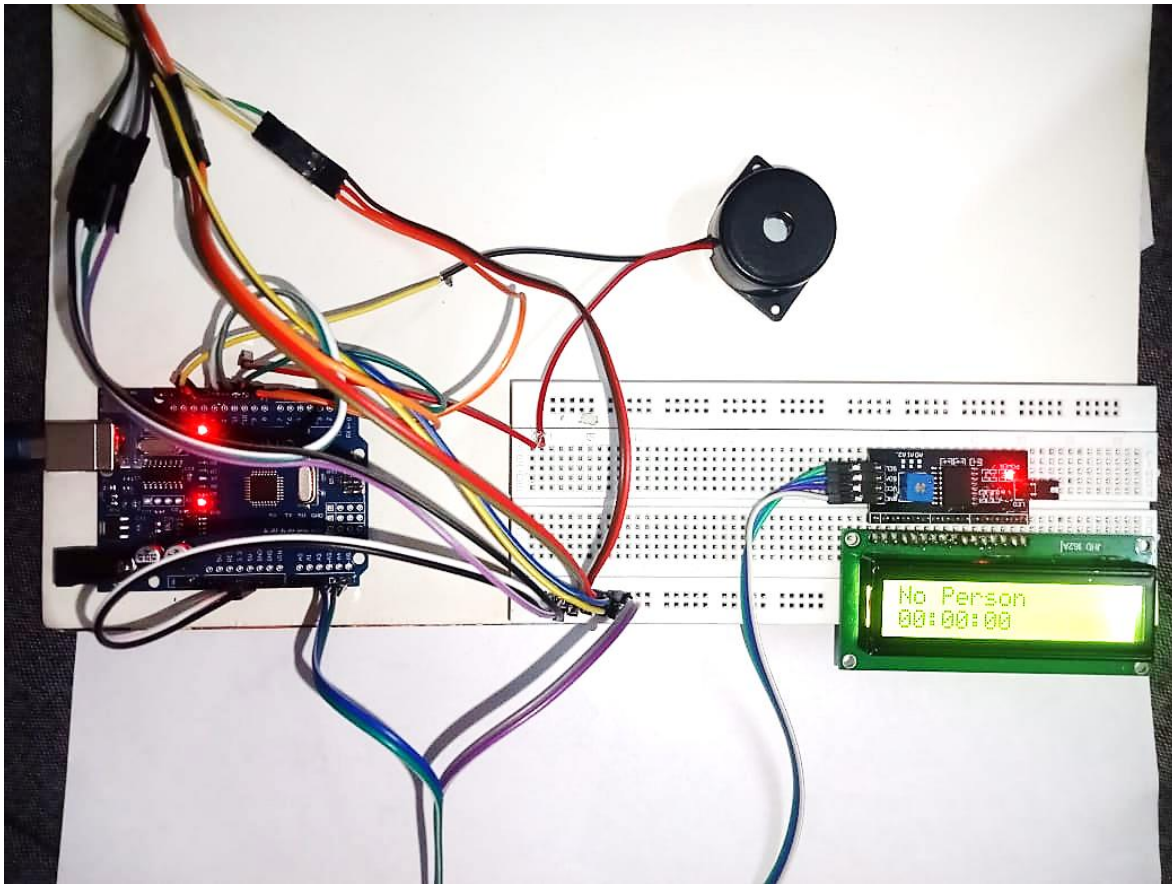


Fig., 10 Working Prototype(Circuit Connections)

IX. COMPLETE CODE

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);

// ===== PINS =====
int pir = 8;
int trigPin = 9;
int echoPin = 10;

int ldr1 = A0; // Ambient light
int ldr2 = A1; // Screen light

int buzzer = 11;

// ===== STATES =====
bool personPresent = false;
bool timerRunning = false;

// ===== TIMERS =====
unsigned long startTime = 0;
```

```

unsigned long lastMotionTime = 0;

// ===== STABILITY CONTROL =====
int lastDistance = 0;
int leaveCounter = 0;
const int leaveConfirmLimit = 5;

// ===== CONSTANTS =====
int safeDistance = 30;
int distanceThreshold = 10;
int lightThreshold = 80;

unsigned long holdTime = 8000;
unsigned long leaveTimeout = 20000;

// ===== SETUP =====
void setup() {
    pinMode(pir, INPUT);
    pinMode(trigPin, OUTPUT);
    pinMode(echoPin, INPUT);
    pinMode(buzzer, OUTPUT);

    lcd.init();
    lcd.backlight();

    showMessage("Smart System", "Starting...");
    delay(2000);
    lcd.clear();
}

// ===== LOOP =====
void loop() {

    bool detected = isPersonPresent();

    if (detected) {

        leaveCounter = 0;
        personPresent = true;

        startTimer();
        displayTime();

        int distance = getDistanceFiltered();
        int lightState = checkBrightness();

        if (distance < safeDistance) {
            lcd.setCursor(0, 1);
            lcd.print("Too Close      ");
            buzzerFast();
        }
        else if (lightState == 1) {
            lcd.setCursor(0, 1);
            lcd.print("Low Brightness ");
            buzzerSlow();
        }
    }
}

```

```

    else if (lightState == 2) {
        lcd.setCursor(0, 1);
        lcd.print("High Brightness");
        buzzerSlow();
    }
    else {
        lcd.setCursor(0, 1);
        lcd.print("All Good      ");
        noTone(buzzer);
    }
}

else {

    if (personPresent) {
        leaveCounter++;

        if (leaveCounter >= leaveConfirmLimit &&
            millis() - lastMotionTime > leaveTimeout) {
            resetSystem();
        }
    }
}

delay(150);
}

// ===== PRESENCE DETECTION (STABLE LOGIC) =====
bool isPersonPresent() {

    int motion = digitalRead(pir);
    int currentDistance = getDistanceFiltered();

    if (currentDistance > 200 || currentDistance <= 0) {
        return false;
    }

    if (motion == HIGH || abs(currentDistance - lastDistance) > 4) {
        lastMotionTime = millis();
        lastDistance = currentDistance;
        return true;
    }

    if (millis() - lastMotionTime < holdTime) {
        return true;
    }

    if (currentDistance < 150 &&
        abs(currentDistance - lastDistance) <= distanceThreshold) {
        return true;
    }

    return false;
}

// ===== ULTRASONIC FILTER =====

```

```

int getDistanceFiltered() {

    int d1 = getDistance();
    delay(10);
    int d2 = getDistance();
    delay(10);
    int d3 = getDistance();

    int arr[3] = {d1, d2, d3};

    for (int i = 0; i < 2; i++) {
        for (int j = i + 1; j < 3; j++) {
            if (arr[i] > arr[j]) {
                int t = arr[i];
                arr[i] = arr[j];
                arr[j] = t;
            }
        }
    }

    return arr[1];
}

// ===== DISTANCE =====
int getDistance() {

    long duration;

    digitalWrite(trigPin, LOW);
    delayMicroseconds(2);

    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);
    digitalWrite(trigPin, LOW);

    duration = pulseIn(echoPin, HIGH, 20000);

    if (duration == 0) return 400;

    return duration * 0.034 / 2;
}

// ===== LIGHT CHECK =====
int checkBrightness() {

    int ambient = analogRead(ldr1);
    int screen = analogRead(ldr2);

    if (screen < ambient - lightThreshold) return 1;
    if (screen > ambient + lightThreshold) return 2;

    return 0;
}

// ===== TIMER =====
void startTimer() {

```

```

    if (!timerRunning) {
        startTime = millis();
        timerRunning = true;
        lcd.clear();
    }
}

// ===== DISPLAY TIMER =====
void displayTime() {

    unsigned long elapsed = (millis() - startTime) / 1000;

    int h = elapsed / 3600;
    int m = (elapsed % 3600) / 60;
    int s = elapsed % 60;

    lcd.setCursor(0, 0);

    if (h < 10) lcd.print("0");
    lcd.print(h); lcd.print(":");

    if (m < 10) lcd.print("0");
    lcd.print(m); lcd.print(":");

    if (s < 10) lcd.print("0");
    lcd.print(s);
}

// ===== BUZZER =====
void buzzerFast() {
    tone(buzzer, 1000);
    delay(50);
    noTone(buzzer);
}

void buzzerSlow() {
    tone(buzzer, 1000);
    delay(150);
    noTone(buzzer);
}

// ===== RESET SYSTEM =====
void resetSystem() {

    personPresent = false;
    timerRunning = false;
    leaveCounter = 0;

    noTone(buzzer);

    showMessage("No Person", "00:00:00");
}

// ===== LCD =====
void showMessage(String l1, String l2) {

```



```

lcd.clear();
lcd.setCursor(0, 0);
lcd.print(11);
lcd.setCursor(0, 1);
lcd.print(12);
}

```

X. RESULT AND DISCUSSION

The performance of the proposed Smart Eye Protection and Alert System was evaluated under different operating conditions. The system responses were observed through the LCD display and buzzer alerts. The following cases demonstrate the functionality of the system.

A. Normal Operating Condition

When all parameters such as distance, brightness, and user activity are within the safe range, the system operates normally without triggering any alert. The LCD displays the message “All Good,” indicating that the user is maintaining proper viewing conditions.



Fig. 11 LCD Output – Normal Condition

B. Unsafe Distance Condition

When all parameters such as distance, brightness, and user activity are within the safe range, the system operates

normally without triggering any alert. The LCD displays the message “All Good,” indicating that the user is maintaining proper viewing conditions.



Fig. 12 LCD Output – Unsafe Distance Condition

C. Low Brightness Condition

When the screen brightness is significantly lower than the ambient light, as detected by the analog LDR sensors, the system identifies this condition as “Brightness Low.” The LCD displays the corresponding message, and an alert is generated to prompt the user to adjust the lighting conditions.



Fig. 13 LCD Output – Brightness Low

D. High Brightness Condition

When the screen brightness is excessively higher than the surrounding light, the system detects this imbalance and classifies it as “Brightness High.” The LCD displays the message “Brightness High,” and a buzzer alert is triggered to indicate the need for adjustment.



Fig. 14 LCD Output – Brightness High

E. Sleep / Inactivity Detection

When no significant movement or change in distance is detected for a predefined duration of ten minutes, the system assumes user inactivity or drowsiness. In this case, the LCD displays the message “Wake Up,” and a continuous buzzer alert is generated to regain the user’s attention.



Fig. 15 LCD Output – Sleep Alert

XI. CONCLUSION

The Smart Eye Protection and Alert System presented in this paper provides an effective solution for monitoring user behaviour and environmental conditions during screen usage. The system successfully integrates multiple sensors, including a PIR sensor, ultrasonic sensor, and analog LDR-based brightness detection, with an Arduino Uno microcontroller to ensure real-time data processing and feedback.

The system is capable of detecting unsafe viewing distance, improper lighting conditions, and prolonged inactivity, which are major factors contributing to eye strain and fatigue. Based on these conditions, it generates appropriate visual and audio alerts, enabling users to take corrective actions immediately. The implementation of brightness classification into “Brightness Low” and “Brightness High” enhances the accuracy of feedback, while the sleep detection feature improves user awareness regarding prolonged inactivity.

The results obtained from both simulation and hardware implementation demonstrate that the system performs reliably and efficiently under various conditions. The integration of multiple sensing techniques improves the overall accuracy and overcomes the limitations of individual

sensors. Additionally, the use of simple and cost-effective components makes the system suitable for practical applications in educational, professional, and home environments.

In conclusion, the proposed system provides a comprehensive and user-friendly approach to promoting healthy screen usage habits. It contributes to reducing eye strain, improving visual comfort, and enhancing overall user well-being. The system also serves as a foundation for further advancements in smart health monitoring and assistive technologies.

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